

OOD detection based on data depth

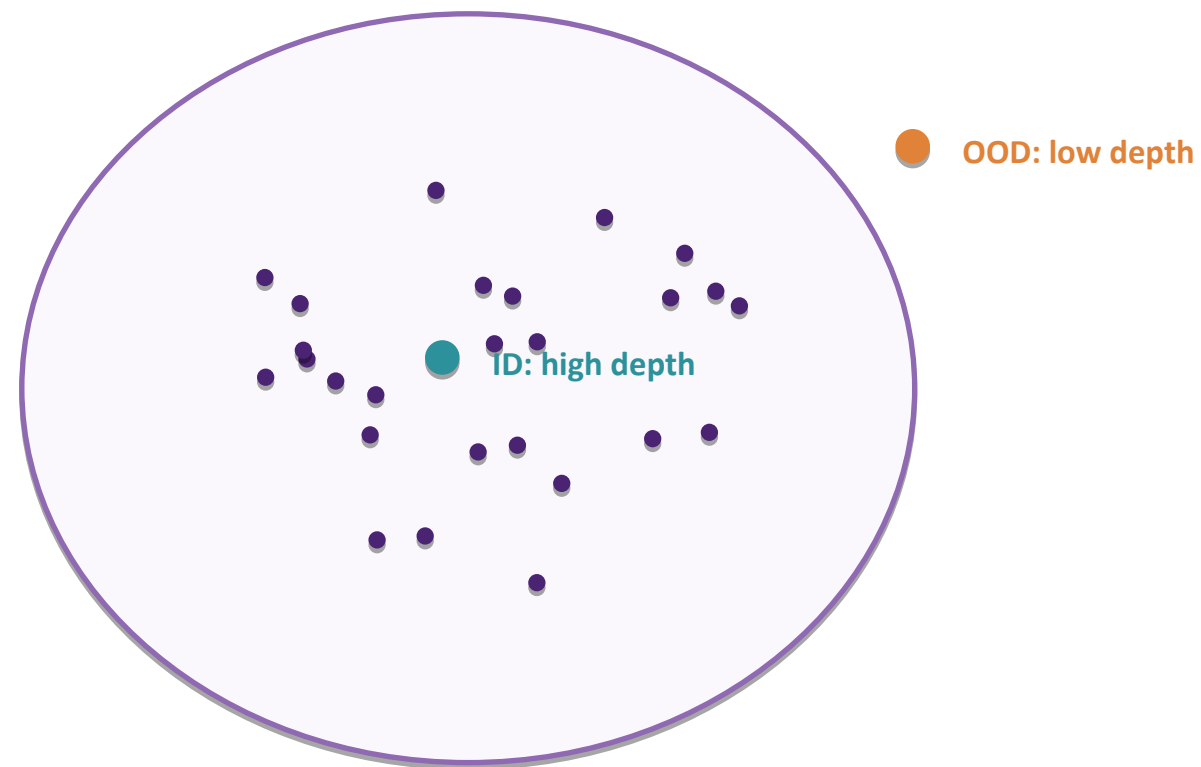
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The problem

Deep models can be highly confident on unfamiliar inputs.

Existing feature methods often use coordinate-wise thresholds or one global distance.

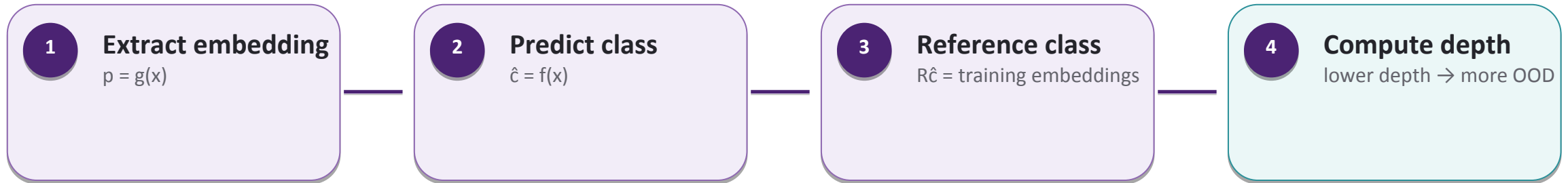
These can miss correlations and class-specific shape and scale.



Key idea: measure how central a test embedding is relative to the training embeddings of its predicted class.

Class-conditional projection depth

A geometry-aware OOD score in the learned feature space



Projection depth = $1 / (1 + \text{worst standardized deviation over all directions})$

Location and scale are estimated robustly with the median and median absolute deviation (MAD).

Why this helps: it evaluates multiple directions, so it can reflect class shape, spread, and cross-dimensional structure.

Computation: the supremum is approximated with refined random search.

Experiments: strong false-positive reduction

ResNet-18 trained on CIFAR-10 · OOD: SVHN, LSUN, Textures

Method	SVHN	LSUN	Textures
MSP	90.39 / 59.83	88.30 / 62.67	88.63 / 61.60
Energy	91.78 / 41.16	89.35 / 48.15	87.73 / 51.01
ViM	91.84 / 41.28	89.39 / 48.27	87.93 / 50.80
DICE	92.65 / 37.53	87.23 / 50.74	85.67 / 56.70
KNN	89.61 / 35.64	89.06 / 41.24	92.41 / 25.24
ProjDepth	89.81 / 33.63	90.24 / 32.36	89.88 / 29.13

Each cell: AUROC / FPR95 (%) · Higher AUROC and lower FPR95 are better.

Key result

ProjDepth gives the lowest FPR95 on SVHN (33.63%) and LSUN (32.36%).

It keeps competitive AUROC across all three datasets.

No method outperforms the others on all datasets.

Takeaway & next step

Projection depth provides a class-conditional, geometry-aware OOD score.

On CIFAR-10, it reduces false positives while maintaining strong AUROC.

Next: scale to ImageNet with ResNet-50 and Vision Transformer embeddings.

Main challenge: when the embedding dimension exceeds the number of samples per class, the available class geometry lies in a lower-dimensional subspace.

Future direction: nonlinear oversampling and depth evaluation for high-dimensional settings.

Thank you